

Google's Project Ara vs ZTE Eco Mobius

Modular smartphones do not seem very far in future. Companies like Google and ZTE have already begun addressing the challenges, and both of them have unveiled their prototypes with intermediate solutions for the issues.

Both the prototypes focus on parts up-gradation and replacement without discarding rest of the phone. While ZTE's Eco Mobius seems to be a quasi-modular architecture with four swappable modules, namely, screen, battery, camera and a combination of processor and memory, Google's Project Ara plans to have more modularity in terms of number and types of modules.

Eco-Mobius architecture doesn't allow adding new kinds of components but Ara's architecture would let you add new type of components, taking the smartphone functionality to a whole new level. Google's proposed device will not only have the basic smartphone modules but also be able to plug modules, like a prototype infra-red imaging lens module for night photography, and even the type of modules which would let you read your blood oxygen levels with a swipe of your finger.

ZTE hasn't revealed much on their design but Google has revealed its complete roadmap. Project Ara is 'open' for all who want to have their say in designing next-generation smartphones for the whole world. Google has created a module development kit, and the module development is open to potentially unlimited number of interested parties. This certainly would help in innovation, accelerating the development and adoption of newer technologies.

What we should look for is an interfacing which could connect modules without snaps or hinges and help make modules with as less added complexity, cost and weight as possible

critical line connecting processor to the memory."

Swapping those modules

A modular phone presents the idea of swapping different modules and configuring the phone features by ourselves. But how can we really swap a camera module with a battery module, or a memory module with a Wi-Fi module?

Actually, the solution is pretty straightforward but it does have some considerations. For such a system, you would have to define a standard interconnect architecture and make the modules intelligent so that they know who they are talking to. Making each module 'smart' or intelligent will not be much of a challenge as we have many low-power and optimally-powerful processors available to do the job.

This way the base module or the main master or the controlling module can easily make decisions on what kind of module is attached to it; it can be a camera or a battery or any other

module. Now everything will only depend on how much area we want to use and what kind of functionality we want to have in the phone.

Some people suggest that we can also have architecture of common control bus and have the modules complying to the control architecture, with a separate data bus which takes care of data. Ashwin Ramachandra, Sasken Communication Technologies, explains, "Looking it from signalling perspective, you need to start looking at signal flows and then identify control signals. If you are able to create a backplane where all your signals flow on one particular set of lines, you are beginning to establish something like that."

We can actually borrow some mechanisms from existing technologies as well. Ashwin adds, "If you go to the automotive segment, there is something called a CAN bus. Let's say, if you start loading your operating system and a signal goes on to your CAN bus, the CAN bus has a mechanism of arbitrating and then trying to figure

out how to start communicating with the signal."

However, such solutions will also carry with them complexity issues.

Power concerns

As already discussed, by making the phones modular one thing we are doing is, having an architecture where we are adding connectors (or interfaces), and in the process of adding these we are adding a lot of communication chipsets. This is going to make a significant impact on the power requirements of the device.

Also, if we are making the modules intelligent enough to be swappable, every module will have a computing element on it, and instead of one computing element now we will have multiple controllers, which means additional power requirement.

Increasing the amount of intelligence on the device will only add to power requirements and power needs will increase in proportion to the number of modules we have on the device.

If we just have a single physical PCB, the shortest route is much smaller. But if you are drawing wires across components, obviously we are taking a longer route. Pure physics tells us that we need extra assistance if we are taking up such a design, which directly affects the performance as a power consumption issue.

RF behaviour

RF behaviour is one of the most significant aspects of regulatory approval for any device. While the industry is still waiting for a functional prototype to test RF behavior on a modular device, many experts predict that the modular architecture would show better RF response.

Nate Srinath, founder-director, Inxece, says, "RF behaviour should have no impact here, or in-fact in modular architecture it will be even better. In our experience we have seen that the more we isolate RF interfaces, the performance gets better. It happens so because when we have the integrated